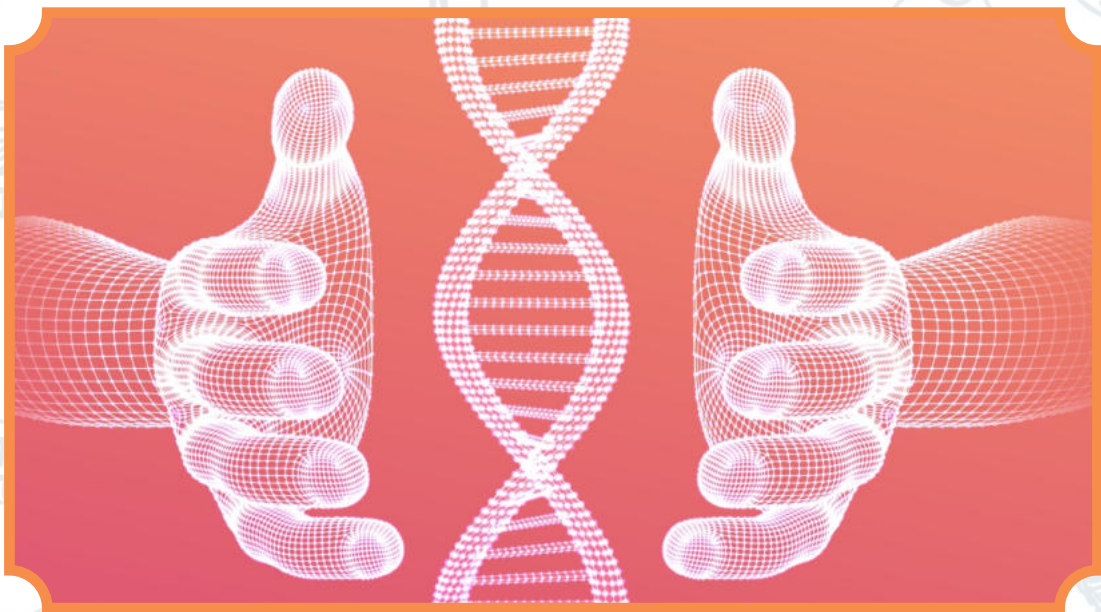


UNIT

22

CHROMOSOME & DNA

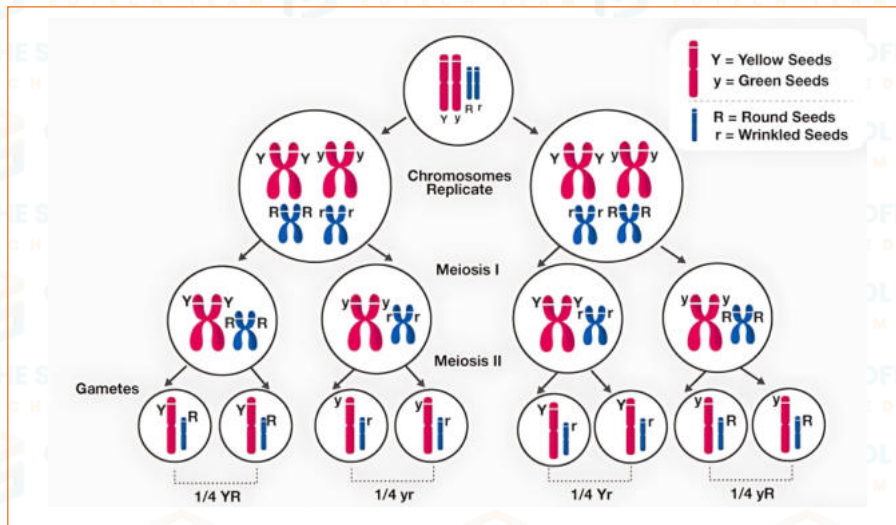




Chromosomal Theory of Inheritance

Background:

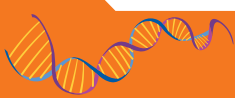
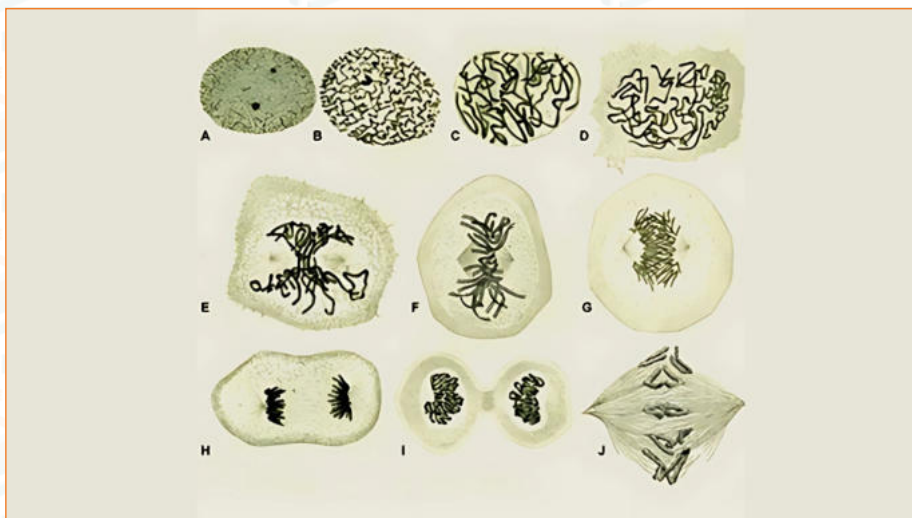
- **Gregor Mendel (1860s):** Proposed laws of inheritance without knowledge of chromosomes.
- **Recognition of Mendel's work (1900):** Rediscovered by **Hugo de Vries, Carl Correns, and Erich von Tschermak.**
- **Cytological Studies (Late 1800s–Early 1900s):** Progress in understanding chromosomes led to integration of Mendelian genetics with cytology.



History of the Chromosomal Theory

1. Walter Fleming (1882):

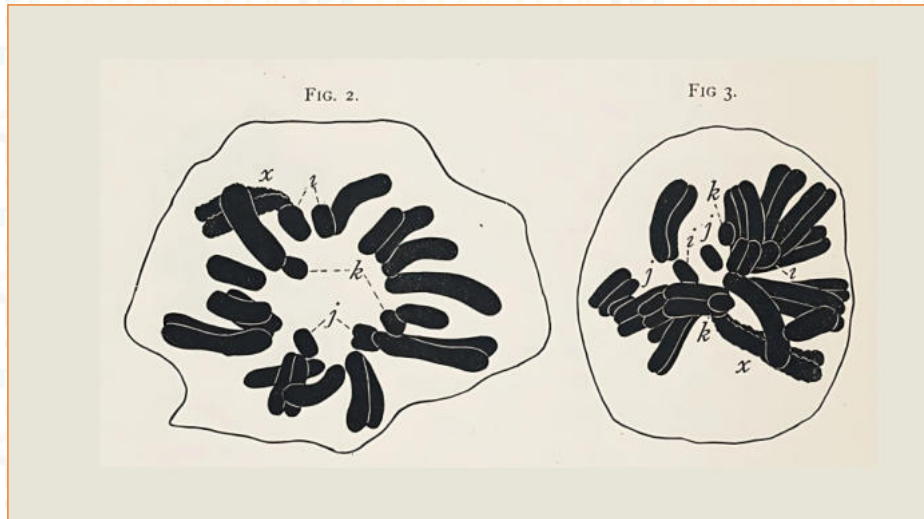
- First observed thread-like, stained structures in dividing cells (salamander larvae) termed **chromosomes**.





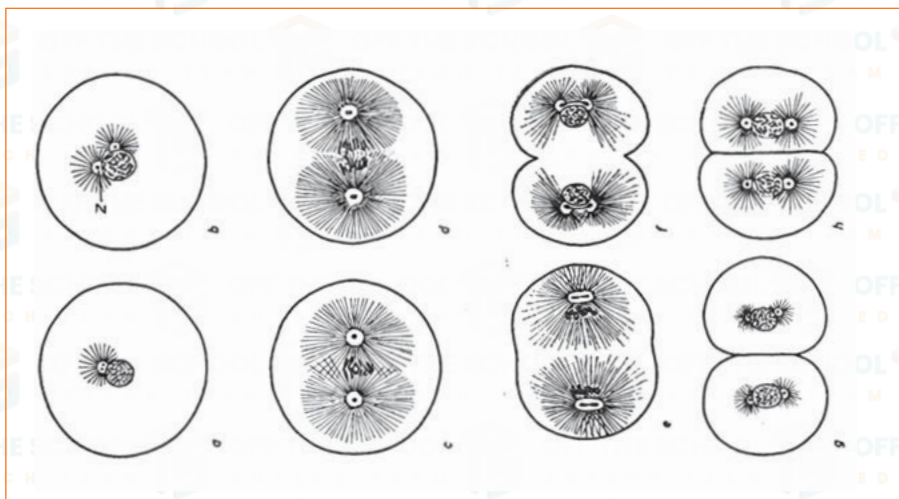
2. Walter Sutton (1902):

- Studied grasshopper cells.
- Noted **chromosome behavior during meiosis** mirrored **Mendelian gene segregation**.
- Published findings in "*The Chromosome in Heredity*".



3. Theodor Boveri:

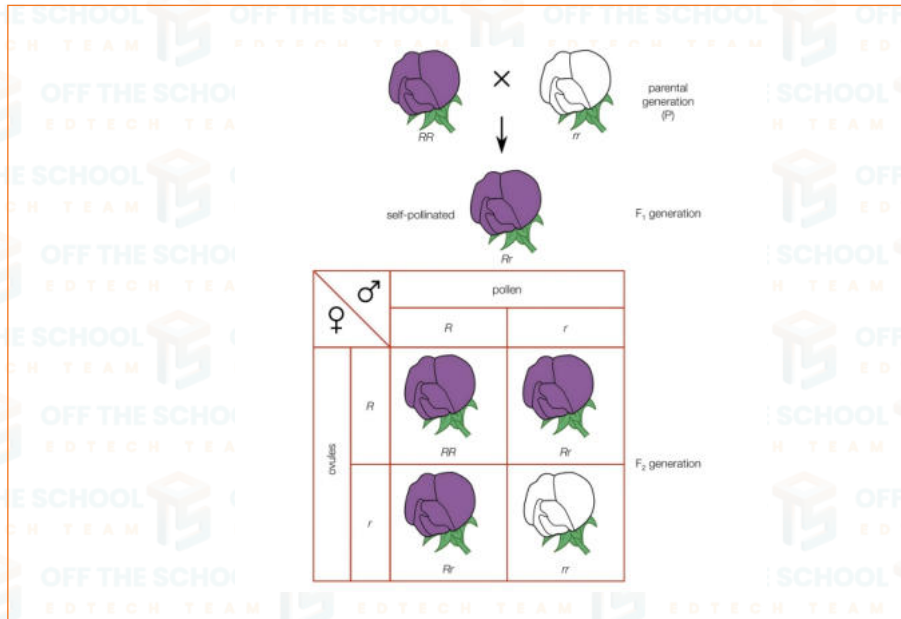
- Observed chromosome number **halves** during meiosis.
- Concluded **egg and sperm carry one set** of chromosomes each.



4. Carl Correns (1900):

- Verified Mendel's work using a different organism.
- Linked **hereditary factors** with **chromosomes**.
- Published paper titled "*Mendel's Law concerning the behavior of racial progeny of hybrids*".





Chromosomal Theory of Inheritance – Statement (W. Sutton, 1902):

- Genes are located on chromosomes.
- Chromosomes are the carriers of heredity.
- Inheritance occurs through chromosomes in gametes.

Evidences Supporting the Theory:

1. Equal contribution from gametes:

- Only **egg and sperm** unite during fertilization.
- Sperm has little cytoplasm → hereditary material must be in the **nucleus**.

2. Segregation during meiosis:

- Chromosomes segregate during meiosis similar to **Mendelian factors**.
- Aligns with **Mendel's Law of Segregation**.

3. Gametes vs. Diploid cells:

- Gametes have **one chromosome from each pair**.
- Diploid individuals have **two copies**, as per **Mendelian concept of alleles**.

4. Independent assortment:

- Homologous chromosome pairs align **independently** at metaphase plate.
- Reflects **Mendel's Law of Independent Assortment**.

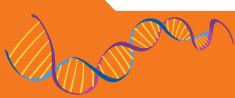
Challenge to the Theory:

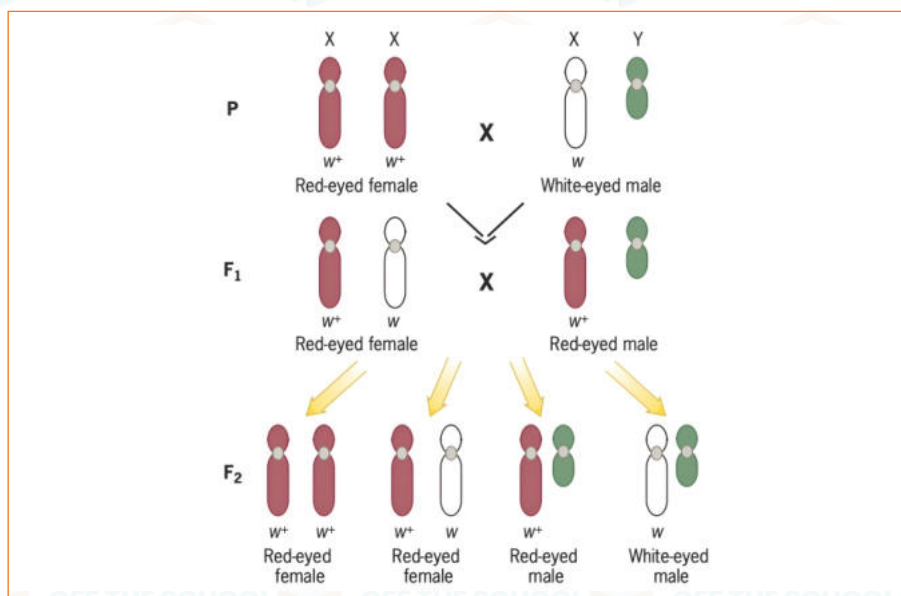
- **Criticism:** Number of independently assorting **Mendelian traits** > **number of chromosome pairs**.

- **Doubt Raised:** If each chromosome carried only one trait, how could so many traits segregate independently?

Resolution (Later Discovered): A chromosome carries **many genes**, and

- **recombination** allows independent assortment even with limited chromosome numbers.





Experimental Verification by T.H. Morgan (1910)

Organism Used:

- **Drosophila melanogaster** (Fruit fly)

- Short life cycle (~2 weeks)
- Easy to maintain
- Large number of offspring
- Only 4 pairs of chromosomes

Initial Cross:

- **Parental (P) Generation:**

- White-eyed **male** × Red-eyed **female**

- **F₁ Generation:**

- All offspring had **red eyes**
- Red eye trait appeared **dominant**

- **F₂ Generation:**

- **Result after F₁ × F₁ cross:**

- ~18% white-eyed flies
- Remaining had red eyes
- **All white-eyed flies were males**

Observation:

- Mendelian ratio was observed **phenotypically**.
- But **only male flies** showed white eyes in F₂ → suggested **sex-linked inheritance**.

Key Questions Raised:

1. Why no white-eyed **females** in F₂?
2. Can females **not** have white eyes?
3. Do white-eyed females **not survive**?





Test Cross:

- **F₁ red-eyed female × white-eyed male** (original P generation)
- **Result:**
 - Offspring included:
 - **White-eyed females**
 - **White-eyed males**
 - **Red-eyed females**
 - **Red-eyed males**
 - **Phenotypic ratio: 1:1:1:1**

Conclusion:

- The **gene for eye color** is located on the **X chromosome**.
- It is a **sex-linked trait**.
- White-eyed females were **possible**, but not produced in earlier cross due to **inheritance pattern**.

Significance of Morgan's Work:

- **First experimental proof** that **genes are located on chromosomes**.
- Supported **Sutton and Boveri's Chromosomal Theory of Inheritance**.
Introduced the concept of **sex-linked inheritance**.

Chromosomes (Chroma = Colour, Soma = Body)

- **Definition:** Literally means "coloured body" – a misnomer, as chromosomes are not inherently coloured.
- **Discovery:**
 - Discovered by **Walter Fleming** in **1882**.
 - Observed in **Salamander larval cells** stained with **Perkin's Aniline dye**.
 - Appeared as **darkly stained thread-like structures**, named **chromosomes**.

Structure and Composition:

- Made of **highly condensed chromatin material**.
- Visible during **cell division**.
- Occur in **species-specific numbers**.
- Carry **genes**.

Sister Chromatids:

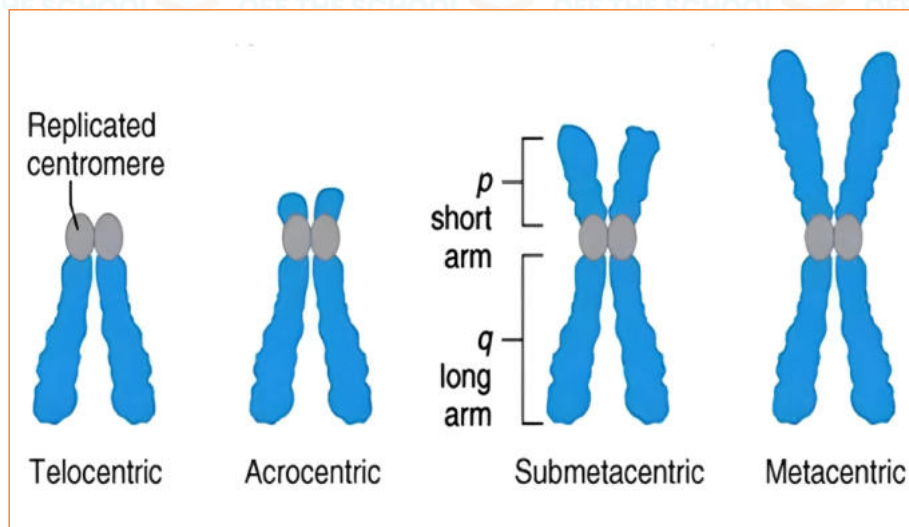
- At the start of cell division, each chromosome consists of **two identical threads** called **sister chromatids**.
- Connected at a **primary constriction** called the **centromere**.
- Each chromatid has **one or two arms**.





Types of Chromosomes (based on centromere position):

1. **Metacentric:**
 - Centromere in the **center**.
 - **Equal arms**.
2. **Submetacentric:**
 - Centromere **slightly off-center**.
 - **Slightly unequal arms**
3. **Sub-telocentric / Acrocentric:**
 - Centromere **far from center**.
 - **One long arm, one short arm**.



Secondary Constriction & Other Features:

- **Secondary constriction:**
 - Also known as **nuclear organizer**.
 - Develops from the **nucleolus** during interphase.
- **Satellite:**
 - **Knob-like end** of chromosome beyond the secondary constriction.
 - Contains **junk DNA**.
- **Telomeres:**
 - **Terminal ends** of chromosomes.
 - Prevent **fusion** between chromosomes.

Chemical Composition of Chromosomes

- **Type:** Chromosomes are **nucleoprotein complexes**.
- **Composition:**
 - **DNA:** ~40%
 - **Protein (mostly histones):** ~60%
 - **RNA:** Present (due to RNA synthesis) but **not a structural component**.





DNA in Chromosomes

- **Structure:**
 - **Highly condensed, double-stranded, and very long.**
 - Continuous and **unbroken along the chromosome.**
- **Length:**
 - In humans, DNA from one set of chromosomes would stretch over **2 meters (~7 feet)** if uncoiled.
 - Yet, it is compacted to fit inside the **microscopic nucleus.**

DNA Packaging Mechanism

1. Nucleosome Formation

- Under **electron microscope**, appears as **"beads on a string"**.
- **200 nucleotides** of DNA coil around a **histone octamer** (core of 8 histone proteins).
- **Histone proteins** are:
 - **Small**
 - Rich in **basic amino acids** (arginine & lysine)
 - **Positively charged**, attract **negatively charged DNA** (due to phosphate).
- This unit is called a **nucleosome**.
- **Linker DNA** (short DNA between nucleosomes) connects them.

Levels of DNA Organization

i) Nucleosome String Formation

- Occurs during **S-phase** after DNA replication.
- DNA wraps around histone core forming **nucleosomes**.
- Nucleosomes are linked by **linker DNA**, forming a **10 nm thick nucleosome string** (beads-on-a-string appearance).

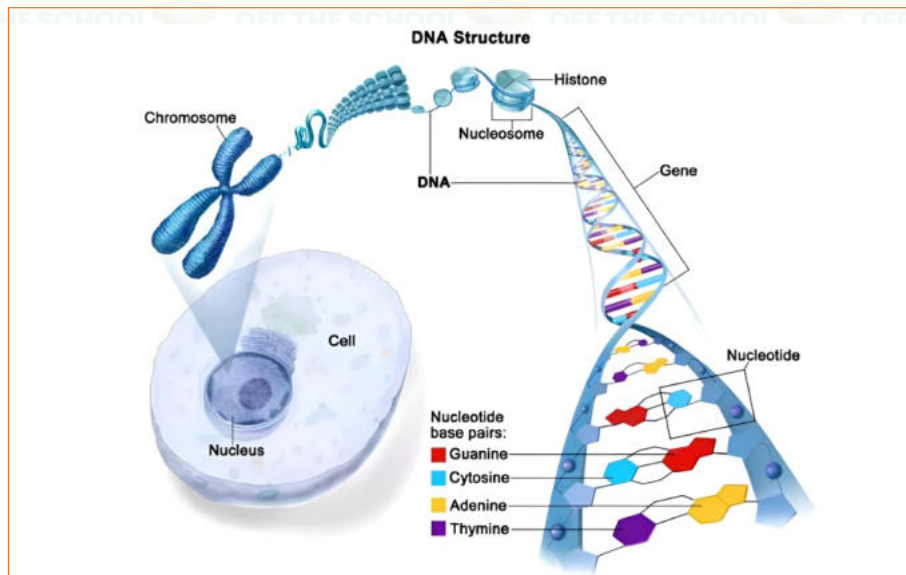
ii) Chromatin Fiber

- Nucleosome string coils further into a **30 nm thick chromatin fiber**.
- Observed during **G2 phase**.
- Two types of chromatin regions:
 - **Heterochromatin:** Highly condensed, **gene-inactive**.
 - **Euchromatin:** Loosely packed, **gene-active**, condenses only during cell division.

iii) Supercoil Formation to Form Chromatids

- In **prophase** of cell division:
 - Chromatin fiber supercoils into **300 nm fibers**
 - These further condense into **chromatids (~700 nm thick)**





Summary of DNA Packaging

STAGE	STRUCTURE FORMED	THICKNESS
DNA wraps histones	Nucleosome	~10 nm
Nucleosomes coil	Chromatin fiber	~30 nm
Supercoiling of chromatin	Chromatid formation	~700 nm

Gene and Gene Locus

Definition of Gene

- **Gregor Mendel** (19th century) introduced the concept of **genes** as "factors" controlling the expression of traits.
- **Gene**: A basic unit of **heredity**, responsible for the inheritance of characteristics.
 - Term "**gene**" was introduced by **William Johansen** in 1909.
 - **Pangene**: Term used by **Bateson** and **Strasburger** for the fundamental unit of heredity in 1906.

Modern Understanding of a Gene

- A **gene** is a **small segment of DNA** that provides instructions for making a specific **polypeptide chain**.
 - These polypeptides function as **enzymes**, helping in various **metabolic reactions** within an organism.
- Genes are composed of **nucleotide sequences**.

Gene Locus

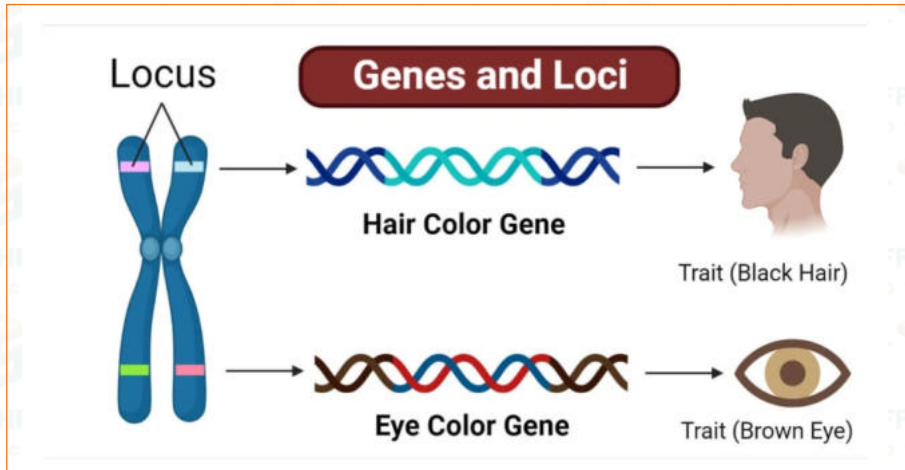
- A **gene locus** (plural: **loci**) is the **specific position** of a gene on a chromosome.
- According to the **chromosomal theory of inheritance**, genes are located on **chromosomes**.





Chromosomes and Homologous/Non-Homologous Chromosomes

- **Diploid cells:** Cells containing **two sets of chromosomes**, one from each parent.
- **Homologous Chromosomes:**
 - Chromosomes that are **morphologically similar** and have the **same gene loci**.
 - Contain **similar genes** (may have the same or different alleles for a trait).
- **Heterologous Chromosomes:**
 - Morphologically **different chromosomes** with **different gene loci**.

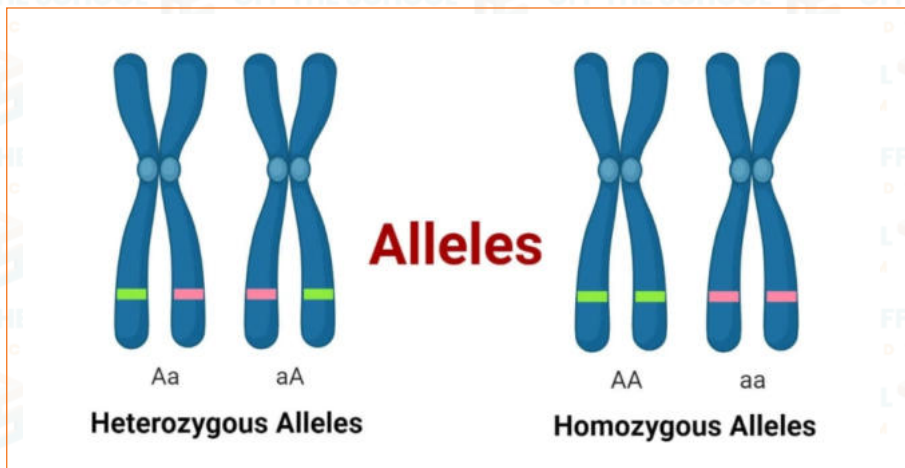


Alleles

- **Alleles:** Different versions (forms) of a gene found at the same gene locus.
 - Example: The gene for **skin color** may have different alleles (e.g., **black** or **white**) located at the same gene locus.
- **Mendel's Findings:**
 - **Two alleles** for each trait (e.g., tall and dwarf plants).
 - **Alternative forms** of a trait controlled by different genes located at the same locus on homologous chromosomes.

Variation Beyond Mendel's Traits

- **Mendelian traits:** Traits controlled by two alleles.
- **Non-Mendelian traits:** Some traits have **more than two alleles** (e.g., blood group).





DNA as Hereditary Material

Chromosomal Theory of Inheritance

- **Chromosomes** contain **DNA and protein**.
- The question arose: Which molecule functions as the **hereditary material** – DNA or proteins, or both?

Fredrick Griffith's Experiment (1928)

- **Organism studied: Streptococcus pneumoniae** (a bacterium causing pneumonia).
 - Two forms:
 - **S-type (smooth, capsulated): Virulent.**
 - **R-type (rough, non-capsulated): Non-virulent.**
- **Griffith's findings:**
 - Injection of **R-type** into mice → no illness.
 - Injection of **S-type** into mice → fatal.
 - **Heat-killed S-type** injected into mice → no ill effects.
 - **R-type + heat-killed S-type** injected → **high mortality**.
- **Conclusion:**
 - R-type bacteria transformed into **S-type** bacteria.
 - This phenomenon was called **transformation**.
 - **Transformation:** A process where genetic material from one bacterium can change the characteristics of another.

Avery, MacLeod, and McCarty (1944)

- **Objective:** Identify the transforming material.
- **Experiment:** Tested various molecules (RNA, protein, DNA) from heat-killed S-type bacteria.
- **Findings:**
 - **DNA** was responsible for the transformation, not RNA or proteins.
 - If **deoxyribonuclease** (which breaks down DNA) was added, the transformation did not occur.
- **Conclusion:** DNA is the **genetic material** responsible for inheritance.

Hershey and Chase Experiment (1952)

- **Objective:** Confirm DNA as hereditary material using **bacteriophage**.
- **Bacteriophage:** A virus that infects bacteria.
 - Composed of **DNA** and **protein**.
- **Procedure:**
 - **Radio-labeling:**
 - **DNA** labeled with **P32** (phosphorus isotope).
 - **Protein coat** labeled with **S35** (sulfur isotope).

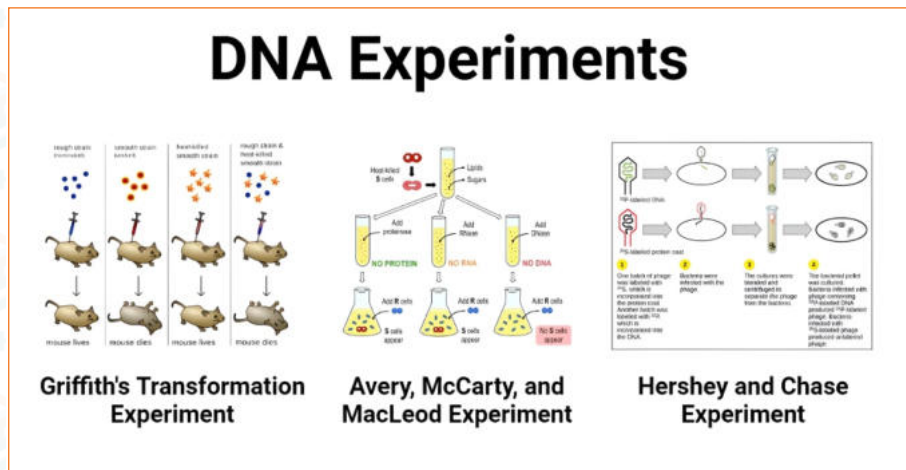




- The **labeled bacteriophage** was allowed to infect **bacteria**.
- After infection, the bacteriophage was allowed to infect the bacteria and **agitated** to separate the outside parts.
- The **pellet** (bacteria) contained **P32-labeled DNA**, but not **S35-labeled protein**.
- **S35** was found in the medium.

Conclusion:

- **Only DNA** from the phage entered the bacterial cell and directed the production of new phages.
- This confirmed that **DNA** is the **hereditary material** in viruses as well.



Summary of Key Experiments

1. **Griffith (1928):** Demonstrated **transformation** in bacteria.
2. **Avery, MacLeod, and McCarty (1944):** Identified **DNA** as the **transforming material**.
3. **Hershey and Chase (1952):** Showed **DNA** as the hereditary material in **bacteriophages**.

DNA Replication

- **Watson and Crick Model of DNA:**
- DNA is a **double helix** with **specific base pairing** between nitrogenous bases.
- The complementary strands of DNA suggest a mechanism for copying genetic material.
- DNA replication is the **exact copying** of DNA that occurs before **cell division** in the **S-phase** of the **cell cycle**.
- **Base-pairing rules:** Each strand serves as a **template** for the synthesis of a new strand.
- **Enzymes** are involved in the process of DNA replication.





Models of DNA Replication

There were three proposed models to explain how DNA replicates:

1. Conservative Model

• Description:

- The **parental DNA** remains **intact** (conserved).
- The **new DNA molecule** is formed entirely from scratch (completely new strands).

• Outcome:

- Parental DNA does not contribute any parts to the new DNA strands.

2. Semi-Conservative Model

• Description:

- The **two strands** of the **parental DNA molecule** separate.
- Each **parental strand** serves as a **template** for the synthesis of a **new complementary strand**.

• Outcome:

- Each **daughter DNA molecule** contains **one old (parental) strand** and **one new strand**.
- Some of the parental DNA is **conserved** in the daughter strands.

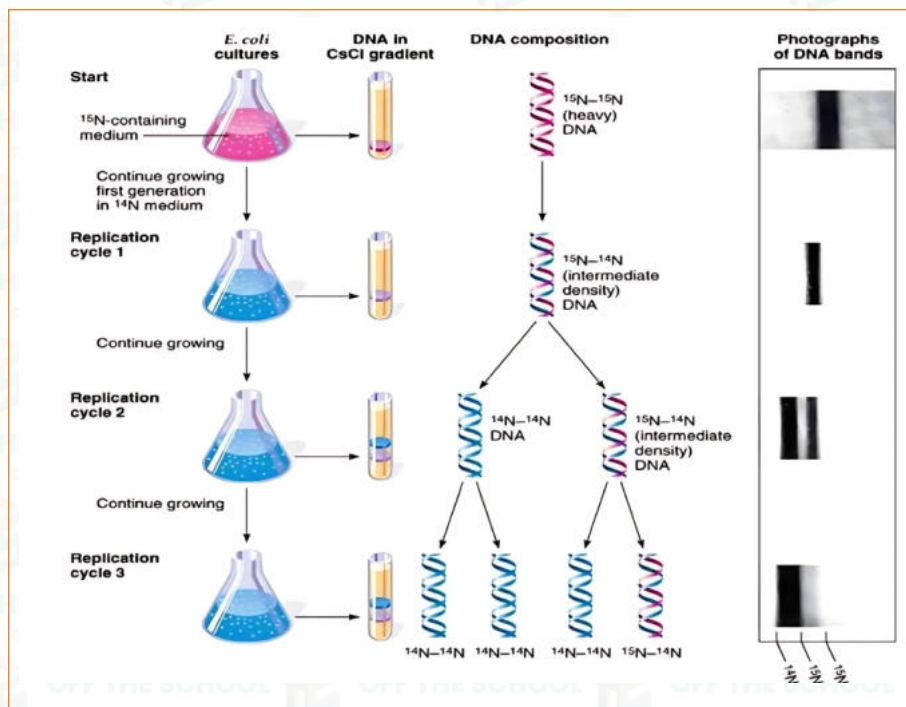
2. Dispersive Model

• Description:

- After replication, **both strands** of the **daughter DNA molecules** are a **mixture of old and new DNA**.

• Outcome:

- No strand is entirely new or entirely old.





Final Model: Semi-Conservative

Experimental Evidence:

- The **semi-conservative model** was confirmed by **Meselson and Stahl's** experiment in 1958, which showed that after one round of replication, each DNA molecule contained one **old** strand and one **new** strand.

Semi-Conservative DNA Replication

- **Watson and Crick's Prediction:**
 - DNA replication is **semi-conservative**:
 - After replication, each **daughter DNA molecule** contains **one old strand** (from the parent) and **one new strand**.
 - The original **double helix** is not fully conserved, but each **strand** of the original duplex is part of the new duplex.

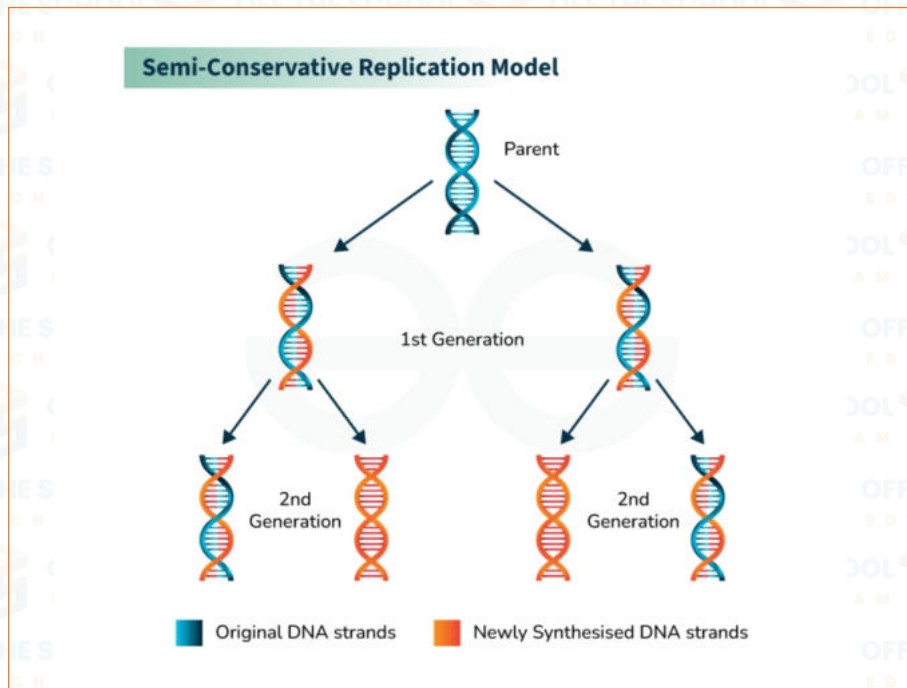
Meselson and Stahl's Experiment (1958)

- **Objective:** Test the semi-conservative model of DNA replication.
- **Method:**
 - **Bacteria (E. coli)** were grown for several generations in a medium containing a **heavy isotope of Nitrogen (N15)**.
 - DNA became **denser** due to incorporation of N15.
 - The bacteria were then transferred to a **medium with lighter Nitrogen (N14)**, and DNA was harvested at different intervals.

Key Findings:

1. **Initial Growth in N15:**
 - The bacteria's DNA was **completely heavy (N15)**.
2. **After First Round of Replication (in N14):**
 - The DNA of the bacteria showed an **intermediate density** between the heavy (N15) and light (N14) isotopes.
 - This indicated that **each new DNA molecule** had **one heavy strand** (from the parent) and **one light strand** (newly synthesized).
3. **After Second Round of Replication (in N14):**
 - Two distinct density classes were observed:
 1. **Intermediate density:** Containing **one heavy strand** and **one light strand**.
 2. **Light density:** Containing **two light strands** (no heavy isotope).
4. **Conclusion:**
 - The experiment confirmed the **semi-conservative model** of DNA replication, where each daughter DNA duplex contains **one strand of old (heavy) DNA** and **one strand of new (light) DNA**.





Summary:

- **Semi-conservative replication** is the process by which **each DNA strand acts as a template** for the synthesis of a new complementary strand.
- **Meselson and Stahl's** experiment provided strong evidence for this model by demonstrating that after replication, **one strand** of each daughter DNA duplex is from the **original DNA**.

DNA Replication Process

DNA replication is a highly accurate and fast process, involving multiple enzymes and proteins. It can be divided into key steps:

(a) Origin of Replication

- **Prokaryotes:** Have a single origin of replication with a specific nucleotide sequence.
- **Eukaryotes:** Have multiple origins of replication on each chromosome.
- **Key Enzymes Involved:**
 - **Helicase:** Unwinds the DNA by breaking the **hydrogen bonds** between complementary bases.
 - **DNA Gyrase:** Works ahead of helicase to relieve the **tension** caused by unwinding.
 - **Single-Stranded Binding Proteins (SSB):** Bind to the separated DNA strands to prevent them from re-joining.
 - The separated DNA forms a **replication bubble** with **replication forks** at each end.





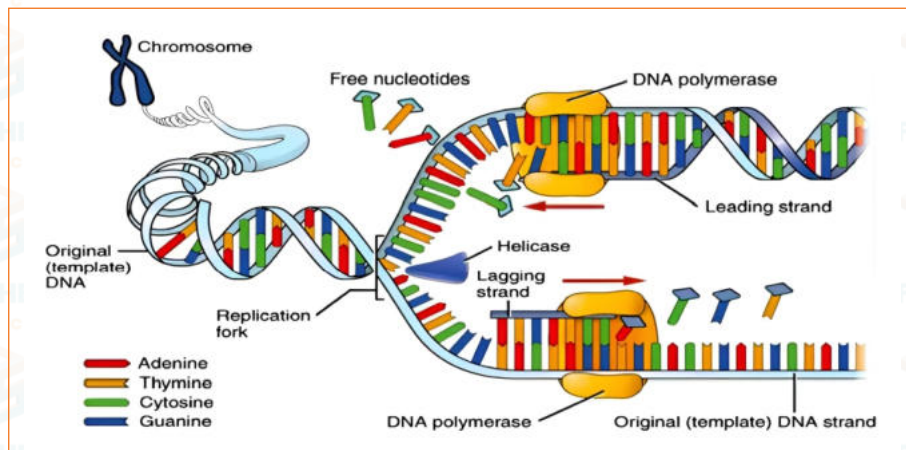
(b) Elongation of New Strands

• DNA Polymerase Enzymes:

- **DNA Polymerase I, II, and III** catalyze DNA elongation.
- Elongation occurs **only in the 5' → 3' direction**.
- **Leading Strand:** DNA polymerase synthesizes continuously in the same direction as the replication fork.
- **Lagging Strand:** DNA polymerase synthesizes in short segments called
- **Okazaki fragments** (100–200 nucleotides in eukaryotes), moving away from the replication fork.
- **Okazaki Fragments** are later joined by **DNA Ligase** to form a continuous strand.

(c) Priming DNA Synthesis

- **Primer:** A short RNA segment (10 nucleotides long) is required to start DNA synthesis.
- **Primase:** An enzyme that synthesizes the RNA primer.
- **DNA Polymerase:** Can only add nucleotides to an existing strand, so a primer is required to start replication.



(d) Types of DNA Polymerase and Their Functions

1. DNA Polymerase I:

- Replaces RNA primers with DNA nucleotides during the **termination phase**.

2. DNA Polymerase II:

- **Proofreading:** Checks for errors and corrects them.
- Repairs DNA damage throughout the cell's lifetime.

3. DNA Polymerase III:

- The **main enzyme** for **elongation**; it synthesizes new daughter strands by adding nucleotides in the 5' → 3' direction.





(e) Termination of Replication

- **Proofreading:** DNA polymerase I removes any wrongly added nucleotides during the elongation phase.
- **Primer Removal:** RNA primers are replaced with DNA nucleotides by **DNA Polymerase I**.
- **Joining of Okazaki Fragments:** DNA ligase **joins** the fragments together to form a continuous strand.

Steps and Enzymes Involved in Replication

1. **Helicase & SSB Proteins:** Unwind the DNA and stabilize single strands.
2. **Primase:** Adds RNA primers to initiate strand synthesis.
3. **DNA Polymerase III:** Synthesizes the leading strand and Okazaki fragments.
4. **DNA Polymerase I:** Replaces RNA primers with DNA nucleotides and proofreads.
5. **DNA Ligase:** Joins Okazaki fragments on the lagging strand.

DNA Replication: Stability and Variability

DNA replication is a critical process that ensures **stability** and **consistency** of genetic information across generations. This process is highly accurate but also allows for occasional **genetic variability** through mutations.

Stability of Genetic Information

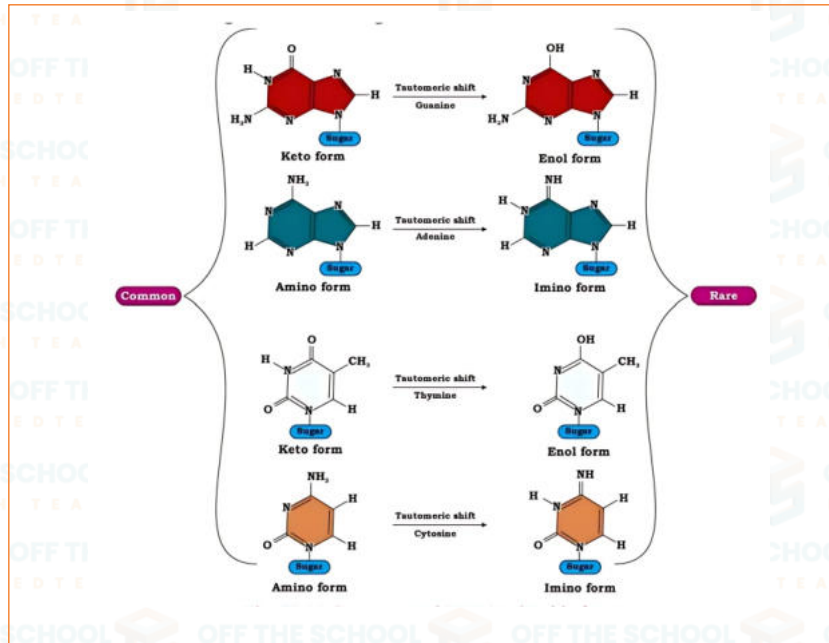
- **Accuracy of DNA Replication:**
 - Replication maintains genetic **stability** by accurately copying DNA, where base pairs of the old strand and newly formed strand form **hydrogen bonds** to guide replication.
 - **Speed of Replication:** DNA polymerase synthesizes 50 to 500 nucleotides per second, and due to this high speed, occasional errors occur.
 - **Error Rate:** In the first step, there is a mistake for every 10,000 base pairs.
 - **Proofreading:** DNA polymerase **III** occasionally incorporates incorrect base pairs, but **DNA polymerase II** proofreads and corrects these errors, maintaining high accuracy.
- **High Fidelity:**
 - **Mammalian Cells**
 - **one mistake per billion base pairs.**
- **Parental and Daughter Strands:** After proofreading, DNA replication results in only about
 - During **prophase**, chromosomes consist of two chromatids, each containing one **parental strand** and one **new daughter strand**.
 - During **anaphase**, the chromatids separate, and each daughter cell receives an exact copy of the genetic material.
- **No Spontaneous Mistakes:** If no errors occur during replication, genetic information remains **constant and stable** from cell to cell and from **parents to offspring**.





Genetic Variability

- **Mutations:** Changes in the DNA sequence can lead to **genetic variability**. A **mutation** is a permanent alteration in the DNA sequence.
 - **De novo mutations** occur when replication errors are not corrected and are passed to the next generation.
- **Replication Timing and Variability:**
 - **Late-replicating DNA** tends to be more prone to mutations.



Sources of Replication Errors

- **Tautomeric Shifts:** One source of replication errors involves the **tautomeric forms** of nitrogenous bases.
 - **Tautomers:** Are interconvertible isomers of bases, where protons shift between different atoms within the base.
 - If a base shifts into its **rare tautomeric form**, it can cause base-pairing mismatches during replication, leading to errors.

	U	C	A	G
U	UUU } Phenylalanine (Phe) UUC } UUA } Leucine (Leu) UUG }	UCU } Serine (Ser) UCC } UCA } UCG }	UAU } Tyrosine (Tyr) UAC } UAA } Stop UAG }	UGU } Cysteine (Cys) UGC } UGA } Stop UGG } Tryptophan (Trp)
C	CUU } Leucine (Leu) CUC } CUA } CUG }	CCU } Proline (Pro) CCC } CCA } CCG }	CAU } Histidine (His) CAC } CAA } Glutamine (Gln) CAG }	CGU } Arginine (Arg) CGC } CGA } CGG }
A	AUU } Isoleucine (Ile) AUC } AUA } Methionine (Met) AUG }	ACU } Threonine (Thr) ACC } ACA } ACG }	AAU } Asparagine (Asn) AAC } AAA } Lysine (Lys) AAG }	AGU } Serine (Ser) AGC } AGA } Arginine (Arg) AGG }
G	GUU } Valine (Val) GUC } GUA } GUG }	GCU } Alanine (Ala) GCC } GCA } GCG }	GAU } Aspartic acid (Asp) GAC } GAA } Glutamic acid (Glu) GAG }	GGU } Glycine (Gly) GGC } GGA } GGG }



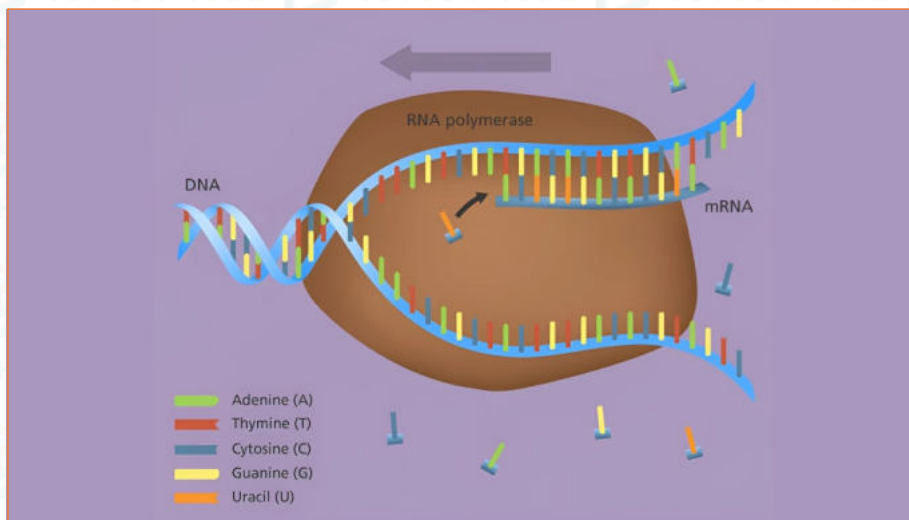


Conclusion

- **Stability:** DNA replication is generally highly accurate, maintaining genetic consistency and stability.
- **Variability:** However, mutations can occur due to errors in replication, which contribute to genetic diversity and can be passed on to future generations.

Gene Expression

Gene expression is the process where the genetic information in a gene is used to synthesize functional products, usually proteins, which are necessary for the survival and functioning of living organisms.



Central Dogma of Gene Expression

The **Central Dogma** describes how genetic information flows from **DNA** to **RNA** and then to **proteins**. It consists of two main processes:

1. Transcription:

- The process of copying information from a specific part of DNA into a complementary form, producing **mRNA (messenger RNA)**.
- The mRNA carries the genetic information from the DNA in the nucleus to the **ribosomes** in the cytoplasm for protein synthesis.

2. Translation:

- The process of converting the information in the mRNA into a sequence of **amino acids** to form a protein.
- This is done with the help of **tRNA (transfer RNA)** and **rRNA (ribosomal RNA)**

Gene and Genetic Code

- A **gene** is a segment of DNA that contains the instructions for synthesizing a protein.
- Different genes have different **base sequences**, leading to proteins with different **amino acid sequences**.
- **Genetic Code:**
 - The genetic code consists of rules that store information for protein synthesis in the DNA.





- DNA and RNA both have four types of nucleotides, but there are 20 amino acids, so the nucleotide sequences cannot directly correspond to amino acids.
- **Triplet Codons:**
 - A **codon** is a triplet of bases that encodes a specific amino acid.
 - There are **64 possible codons**, more than enough to encode all 20 amino acids.
 - **Francis Crick** and colleagues proved that a codon consists of 3 bases.

Genetic Code Table

- **Codons:** A specific set of three nitrogenous bases on mRNA that code for one amino acid.
- There are **61 codons** for 20 amino acids, and **3 stop codons** (UAG, UAA, UGA).
- **Start Codon:** The codon **AUG** signals the start of protein synthesis.
- The genetic code is **redundant** but **not ambiguous** (each codon codes for one amino acid but multiple codons can encode the same amino acid).

Example

- **Codon for Isoleucine:** **AUC** is the codon for Isoleucine.
- **Redundancy:** Several codons can encode the same amino acid (e.g., **6 codons** for Arginine).

Key Points

- The **genetic code** is the language that the cell uses to convert DNA information into proteins.
- **mRNA** carries the information from DNA and uses codons for **amino acid sequence**.
- **Codons** are triplets of bases on the mRNA, and each codon encodes a specific amino acid or a stop signal.

Mechanism of Transcription

Transcription is the process by which a specific segment of DNA is copied into RNA. This process plays a crucial role in gene expression, where the genetic code stored in DNA is transferred to RNA, which is later used for protein synthesis. Transcription has some specific characteristics and limitations:

Limitations of Transcription

1. **Selective Transcription:** Only certain regions (genes) of the DNA are transcribed. For example, the cells in hair follicles only transcribe the DNA sequences that encode for keratin protein synthesis.





2. **Occasional Transcription:** Transcription happens only when needed. It copies one strand of DNA (template strand) because the two strands of DNA are complementary but not identical.

- **Template Strand (Anti-sense strand):** The strand that carries the genetic information used for RNA synthesis.
- **Coding Strand (Sense strand):** The strand that is complementary to the template strand, not directly involved in RNA synthesis.

The enzyme **RNA polymerase** synthesizes RNA in the **5' to 3' direction**.

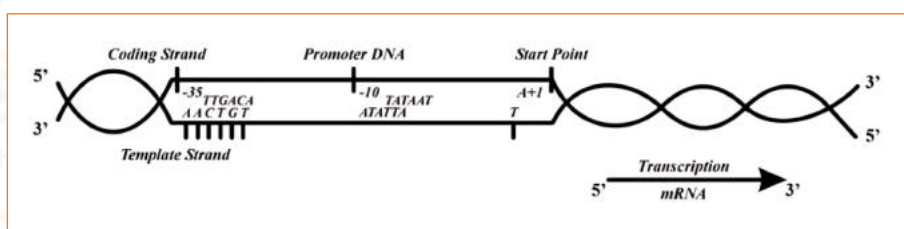
Steps of Transcription

The transcription process can be divided into three main stages:

1. **Initiation**
2. **Elongation**
3. **Termination**

1. Initiation

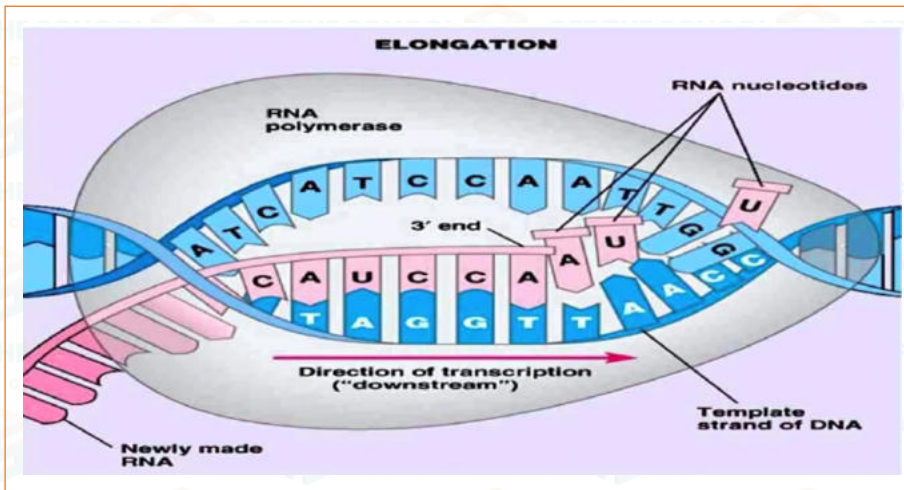
- **Promoter Region:** Transcription begins when **RNA polymerase** binds to a specific sequence of DNA called the **promoter** region, located near the start of a gene.
 - In **prokaryotes**, common promoter sequences are **TATAAT** (-10 sequence) and **TTGACA** (-35 sequence).
 - In **eukaryotes**, the promoter includes sequences like **TATA box** (-25 sequence) and **CAAT box** (-70 sequence).
- **RNA Polymerase Binding:** The RNA polymerase recognizes the promoter sequence and binds to it, starting the transcription process.



2. Elongation

- **DNA Unwinding:** RNA polymerase unwinds the DNA double helix at the start of the gene.
- **RNA Synthesis:** RNA polymerase moves along the template strand of the DNA in the **3' to 5' direction**, synthesizing a complementary RNA strand in the **5' to 3' direction**.
 - The base pairing rules used in RNA synthesis are similar to DNA replication, except **Uracil (U)** pairs with **Adenine (A)** instead of **Thymine (T)**.
 - For example: **A → U, T → A, C → G, G → C**.
 - As the RNA polymerase synthesizes the RNA, the RNA molecule detaches from the DNA after about 10 nucleotides are added. The RNA strand drifts away from the DNA template, elongating continuously.





3. Termination

- **Termination Region:** Transcription continues until RNA polymerase reaches a specific **terminator region** of the gene.
 - The **terminator sequence** is usually made up of **GC base pairs** followed by **AT base pairs**.
- **RNA Detachment:** When the RNA polymerase reaches the terminator region, two events occur:
 - The **RNA molecule** completely detaches from the DNA.
 - The **RNA polymerase** dissociates from the template strand.
- The **GC hairpin** structure in the RNA forms, which causes the RNA polymerase to stop synthesizing RNA.

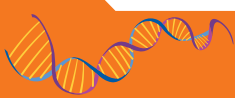
Key Points:

- Transcription is **selective** and occurs **only when needed**.
- RNA polymerase synthesizes RNA from the **template strand** in the **5' to 3' direction**.
- The process involves three stages: **initiation**, **elongation**, and **termination**.
- Transcription is terminated when a **GC hairpin** and **polynucleotide tail** cause the RNA polymerase to stop.

This process results in the formation of **mRNA**, which can be used in **translation** to synthesize proteins.

Modification in mRNA Before Translation

In eukaryotes, the mRNA undergoes several modifications before it can be translated into a protein. These modifications ensure the stability of the mRNA and its proper translation. In contrast, **prokaryotic cells** do not modify their mRNA because both **transcription** and **translation** occur simultaneously in the cytoplasm.





Key Concepts in mRNA Modification (Eukaryotes):

1. Exons and Introns:

- **Exons:** Coding regions of the gene that are translated into proteins.
- **Introns:** Non-coding regions that do not code for proteins and must be removed during mRNA processing.

Steps of mRNA Modification in Eukaryotes:

1. RNA Splicing:

- **RNA splicing** is the process where **introns** are removed from the primary mRNA, and **exons** are joined together to form the final mRNA. This ensures that only the coding regions (exons) are retained.
- RNA splicing requires a complex of proteins and small nuclear RNAs called **snRNPs** (small nuclear ribonucleoproteins).

Steps of RNA Splicing:

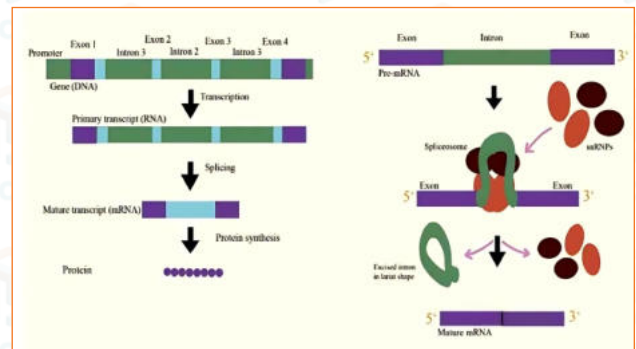
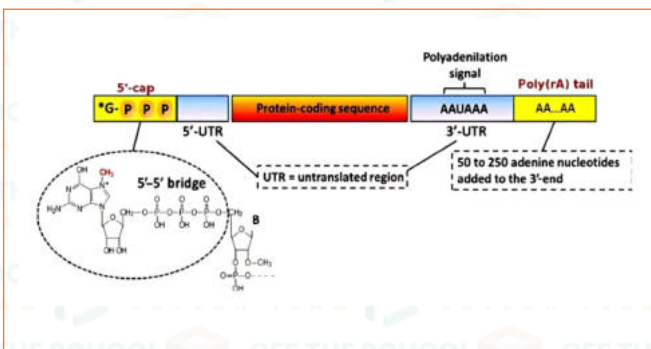
- **Step 1:** **snRNPs** bind to the intron regions of the primary mRNA.
- **Step 2:** The binding of **snRNPs** causes the folding of the mRNA, bringing the **5'** and **3'** ends of the intron closer together, creating a **loop**. This brings the exons closer together as well.
- **Step 3:** The intron is excised, and the exons are joined together to form the mature mRNA. The excised introns are used in other cellular processes, and the **snRNPs** are released and reused.

2. Addition of 5' Cap:

- A **7-methyl GTP cap** is added to the **5' end** of the mRNA.
- This cap is crucial for **protecting the mRNA from degradation** and aids in the **initiation of translation**.

3. Addition of Poly-A Tail:

- A **poly-A tail** (a string of adenine nucleotides) is added to the **3' end** of the mRNA. This tail can range from **50 to 250 adenine nucleotides**.
- The poly-A tail helps in **protecting the mRNA from enzymatic degradation** and plays a role in **exporting the mRNA from the nucleus** and **translation initiation**.





Protective Functions of mRNA Modifications:

- The **5' cap** and **poly-A tail** protect the mRNA from degradation by **nucleases** (enzymes that break down nucleic acids).
- These modifications ensure that the mRNA remains **stable** and is efficiently translated into a protein.

Diagram of mRNA Modification:

- **5' Cap** (7-methyl GTP) is attached at the 5' end.
- **Poly-A Tail** is added at the 3' end, consisting of adenine nucleotides.
- **5' UTR** (untranslated region) and **3' UTR** are regions that are transcribed but not translated.
- **Coding Sequence**: The region that will be translated into protein.

Summary:

- **mRNA Modification** is essential in eukaryotes to prepare the mRNA for translation.
- **RNA splicing** removes non-coding introns and joins the coding exons.
- **5' Cap** and **Poly-A Tail** are added to protect mRNA from degradation and aid in translation initiation.

Translation: The Second Phase of Gene Expression

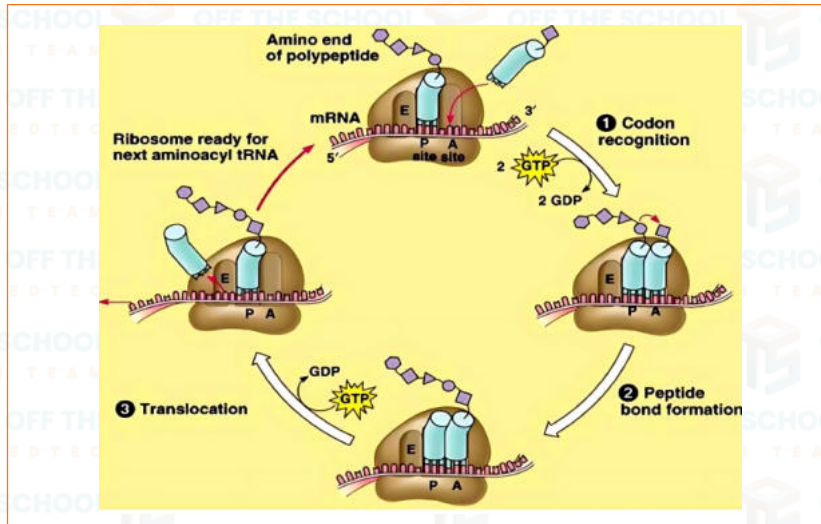
Translation is the process in which **mRNA** is decoded by ribosomes, with the help of **tRNA**, to synthesize a **polypeptide chain** (protein). Translation occurs in four key phases: **Activation of Amino Acids**, **Formation of the Initiation Complex**, **Elongation**, and **Termination**.

Phases of Translation

(i) Activation of Amino Acid

- **Amino acids** are either obtained from food or synthesized by the cell.
- Each amino acid is **activated** by binding to its specific **tRNA**.
- **tRNA Structure**:
 - The **3' end** of tRNA has an **amino acid attachment site**.
 - The **5' end** of tRNA has an **anticodon site** where the **anticodon** (complementary to mRNA codon) binds.
 - An enzyme called **aminoacyl-tRNA synthetase** attaches the amino acid to its respective tRNA.
 - There are **20 different aminoacyl-tRNA synthetases**, one for each amino acid.





(ii) Formation of Initiation Complex

- The **initiation complex** consists of:
 - **Small ribosomal subunit**.
 - **mRNA** (with its 5' cap bound to the ribosome).
 - The first **aminoacyl-tRNA** with **methionine** (in eukaryotes) or **N-formyl methionine** (in prokaryotes).
- **Initiation factors** help assemble the initiation complex:
 - **Initiation factor-I** helps in the formation of the complex.
 - **Initiation factor-II** helps the mRNA bind to the small ribosomal subunit.
- The **large ribosomal subunit** then attaches to the small subunit, forming the complete **ribosome**.

(iii) Elongation

- The ribosome has **three sites**:
 - **P site** (Peptidyl site) – holds the tRNA with the growing polypeptide.
 - **A site** (Aminoacyl site) – holds the tRNA with the next amino acid.
 - **E site** (Exit site) – holds the tRNA before it exits the ribosome.
- The process of **elongation** proceeds as follows:
 - A **new tRNA** with the correct **amino acid** binds to the **A site**.
 - The **peptidyl transferase** enzyme forms a **peptide bond** between the amino acids at the **P** and **A sites**.
 - The ribosome then **translocates**, shifting along the mRNA from the **5'** to the **3'** direction.
 - The empty tRNA at the **P site** shifts to the **E site** and exits, while the tRNA at the **A site** moves to the **P site**.
 - This process repeats until a **stop codon** is reached.

(iv) Termination

- **Termination** occurs when the ribosome encounters a **stop codon** (such as UAA, UAG, or UGA) at the **A site**.





- Stop codons do not bind with any tRNA, which halts the elongation process.
- The **polypeptide chain** is then released from the ribosome.
- The **ribosomal subunits** dissociate from the mRNA, and translation is complete.

Comparison: Protein Synthesis in Prokaryotes vs. Eukaryotes

FEATURES	PROKARYOTES	EUKARYOTES
Ribosomes	Ribosomes (70S) are in the cytoplasm.	Ribosomes (80S) are attached to the RER
Gene Location	Genes are in the cytoplasm, so transcription and translation occur together.	Genes are in the nucleus , so transcription occurs in the nucleus and translation in the cytoplasm.
Introns	Genes do not contain introns.	Genes contain introns and exons
Promoter Sequences	Promoters are TATAAT (-10) and TTGACA (-35)	Promoters are TATA box (-25) and CAAT box (-70) .
Initial Amino Acid	Uses N-formyl methionine	Uses methionine
mRNA Modifications	No 5' cap or poly-A tail modifications.	mRNA is modified with a 7-methyl GTP cap at the 5' end and a poly-A tail at the 3' end.

Summary of Translation Process

- **Activation of amino acids** involves attaching amino acids to their corresponding tRNA molecules.
- **Initiation** forms the initiation complex with ribosomes and tRNA.
- **Elongation** adds amino acids to the growing polypeptide chain.
- **Termination** occurs when a stop codon is reached, and the polypeptide chain is released.

Importance of Regulating Gene Expression

1. Energy Conservation and Space Efficiency:

Gene expression is an energy-intensive process, and regulating it ensures that genes are only expressed when needed, conserving energy. Additionally, DNA is highly coiled in the nucleus, and only the relevant portion of DNA unwinds for gene expression, saving space within the cell.





2. **Cellular Differentiation in Multicellular Organisms:**

In multicellular organisms, regulation of gene expression allows cells to specialize. Even though all cells contain the same genome, only specific genes are expressed in each cell type, leading to differentiation and the formation of specialized tissues.

3. **Regulation of Metabolism:**

Gene expression regulation is crucial for maintaining normal metabolism. For example, liver cells express the gene for alcohol dehydrogenase to break down alcohol, but red blood cells do not because they lack the necessary gene expression for this enzyme.

4. **Prevention of Metabolic and Genetic Disorders:**

When gene expression regulation is disrupted, it can lead to diseases such as cancer or metabolic disorders. Proper regulation is vital for the proper functioning and health of organisms.

Methods of Gene Regulation

1. **Positive Gene Regulation:**

Gene expression is activated by specific proteins known as activators. This type of regulation helps in turning on the gene expression when needed.

2. **Negative Gene Regulation:**

Gene expression is suppressed by repressor proteins, which prevent the gene from being expressed.

3. **Gene Regulation in Unicellular Organisms:**

Unicellular organisms like *E. coli* regulate their gene expression to adapt to environmental changes. For example, when lactose is present, enzymes are synthesized to metabolize it. When lactose is absent and another sugar is present, those enzymes are repressed.

4. **Gene Regulation in Multicellular Organisms:**

In multicellular organisms, gene expression is tightly regulated to ensure that specific genes are expressed in the right tissues. This regulation also plays a crucial role in development, as it controls the formation of different tissues and cell types in the embryo.

Levels of Gene Regulation

1. **Epigenetic Level:**

Regulation occurs when DNA is uncoiled from the nucleosome to bind with transcription factors, allowing gene expression.

2. **Transcriptional Level:**

Regulation occurs during the transcription process, influencing which genes are transcribed into RNA.





3. **Post-transcriptional Level:**

Regulation occurs after transcription, during the transport of RNA from the nucleus to the cytoplasm, or during RNA processing (like splicing).

4. **Translational Level:**

Gene expression can be regulated during translation, when mRNA is used to synthesize proteins.

5. **Post-translational Level:**

Regulation can occur after proteins are synthesized, influencing their activity, folding, and function.

Role of Introns and Exons in Gene Expression

1. **Exons and Introns:**

In eukaryotes, genes contain both coding (exons) and non-coding (introns) regions. Exons are expressed in the final protein, while introns are removed during RNA splicing before translation.

2. **Alternative Splicing:**

Introns play a crucial role in alternative splicing, a process that allows a single gene to produce multiple protein variants. This increases the protein diversity in the cell.

3. **Function of Introns:**

Despite being non-coding, introns are not "junk." They are essential for:

- Enhancing protein diversity through alternative splicing.
- Regulating gene expression positively.
- Involvement in transcription initiation, termination, and mRNA transport.
- Influencing chromatin assembly.
- Acting as signals for transcription initiation, particularly the first intron.

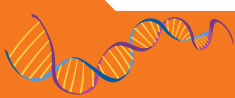
4. **Intron-Related Research:**

Ongoing research continues to uncover the multiple roles of introns in gene expression regulation, including their importance in transcription and their association with chromatin structure.

Mutation: Overview and Types

Definition of Mutation

- Mutation refers to changes in the DNA that cause errors in the transmission of genetic information. It is the failure to store or replicate genetic information faithfully.
- In eukaryotes, DNA is located on chromosomes, and mutations can affect both the structure and number of chromosomes or changes within individual genes.





Sources of Mutation

1. Mutagens:

- Mutagens are substances that cause mutations. These can be **physical**, **chemical**, or **biological** in nature.

2. Physical Mutagens:

- These are physical forces that can damage the DNA. They break bonds in the DNA, such as the phosphodiester bond or the bonds between nitrogenous bases.
- Examples include:
 - Ionizing radiation (gamma rays, X-rays)
 - Non-ionizing radiation (ultraviolet radiation)
 - High temperature
 - Cosmic waves
 - Ultrasonic waves

3. Chemical Mutagens:

- Certain chemicals can alter the structure of DNA or replace nucleotides. These include environmental or industrial chemicals.
- Examples:
 - Nitrous oxide
 - Mustard gas
 - Photochemicals
 - Radioactive isotopes
 - Free oxygen

4. Biological Mutagens:

- Some microorganisms or biochemical agents produced during metabolism can act as mutagens. Additionally, errors in normal physiological processes (e.g., disjunction errors leading to heteroploidy) can cause mutations.
- Example:
 - Some viruses (e.g., HIV) can induce mutations like Kaposi's sarcoma (skin cancer).

Types of Mutations

1. Mutagenesis:

- The process of mutation is known as **mutagenesis**, and organisms affected by mutations are called **mutants**. The organisms that retain their original genetic material are called **wild types**.

2. Classification of Mutations:

- Mutations can be classified into different types based on their impact and origin.





a) Negative Mutation:

- This type of mutation is harmful to the organism and leads to various abnormalities.
- **Examples:**
 - Sickle cell anemia
 - Down's syndrome
- Harmful mutations are often eliminated by **natural selection** from the gene pool.

b) Positive Mutation:

- Positive mutations result in new alleles that offer an adaptive advantage, making organisms more suited to their environment.
- **Examples:**
 - Mutation for black skin color, which provides protection against UV radiation
 - Polyploidy in plants, leading to different varieties
 - Adaptive changes like beak and claw shapes in birds
- These mutations are selected for by **natural selection**.

c) Neutral Mutation:

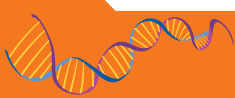
- Neutral mutations do not have a significant effect on the phenotype of the organism. These mutations neither harm nor benefit the organism, and they are inherited in the usual way.
 - These mutations do not cause observable effects on the organism's survival or reproduction.

Role of Mutation in Evolution

- Mutation plays a vital role in providing genetic diversity and **fitness** to adapt to changing environments, driving the **evolutionary process**.
- For example, mutations that allowed aquatic organisms to survive on land were crucial for the evolution of terrestrial animals and plants.
- Organisms with mutations necessary for survival in a new environment are selected by nature, while those without such mutations fail to thrive in that environment.

Conclusion

- Mutation is not always harmful. While negative mutations can lead to diseases, positive mutations contribute to evolutionary adaptations. Neutral mutations have no observable effects but contribute to genetic diversity.
- Mutation is a natural process that facilitates **evolution** and the **adaptation** of organisms to their environments.





Chromosomal Mutation

Chromosomal mutations refer to changes in the number or structure of chromosomes. These mutations can lead to various genetic disorders and are classified into two main types:

1. **Heteroploidy**
2. **Chromosomal Aberration**

1. Heteroploidy

Heteroploidy is a change in the number of chromosomes, often due to errors in chromosome segregation during **meiosis** (non-disjunction). Heteroploidy is further classified into two types:

a) Polyploidy

Polyploidy refers to the condition where there is an increase in the number of chromosome sets.

- **Triploid (3N)** and **Tetraploid (4N)** organisms are examples.
- Common in **plants**, polyploidy leads to varieties within a species (e.g., different varieties of **wheat** and **rice**).
- It can result in larger plant size and more resilient crops.

b) Aneuploidy

- **Aneuploidy** occurs when there is a gain or loss of one or two chromosomes.
 - **Monosomy**: Loss of one chromosome (e.g., $2N - 1$).
 - **Trisomy**: Gain of one chromosome (e.g., $2N + 1$).

Examples of **Aneuploidy** in humans include:

1. **Down's Syndrome** (Trisomy 21):

- **Cause** : Trisomy of chromosome 21 (3 copies instead of 2).
- **Symptoms**: Abnormal body and mental development, round face, small head, short hands, wide fingers, long tongue.
- **Treatment**: No cure; psychological counseling and medication for behavior management.

2. **Klinefelter's Syndrome** (Trisomy of sex chromosomes XXY):

- **Cause**: Extra X chromosome in males ($44 + XXY$).
- **Symptoms**: Feminine characteristics (less body/facial hair, enlarged breasts, wide hips), small testicles, infertility.
- **Treatment**: Testosterone therapy to help with some symptoms but does not cure infertility.

3. **Turner's Syndrome** (Monosomy of sex chromosome X):

- **Cause**: Missing X chromosome in females ($44 + X$).
- **Symptoms**: Short stature, infertility, webbed neck, low IQ.
- **Treatment**: Growth hormone therapy and female hormonal therapy for puberty and reproductive cycle.





2. Chromosomal Aberration

Chromosomal Aberration refers to structural changes in chromosomes, including alterations in the chromosome structure or the position of genes. There are four major types of chromosomal aberrations:

a) Deletion or Deficiency

- **Deletion** occurs when a part of a chromosome breaks off and is lost.
 - This results in a **reduction in the chromosome's size** and can cause the loss of genetic material, leading to various genetic disorders.

b) Duplication

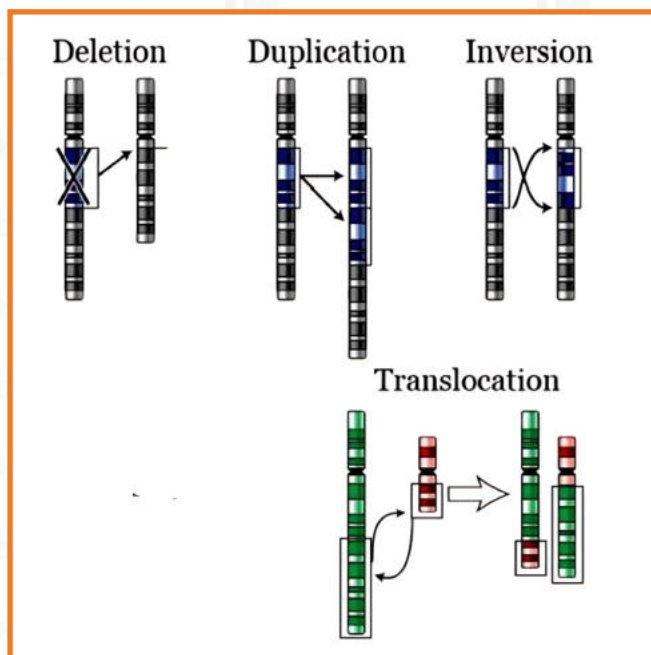
- **Duplication** occurs when a portion of a chromosome is duplicated and attached to its homologous chromosome.
 - This results in **gene duplication** and increases the chromosome's size.
 - Duplication can lead to gene redundancy or altered gene function.

Translocation

- **Translocation** occurs when a part of a chromosome breaks off and attaches to a **non-homologous chromosome**.
 - This can increase the chromosome's size but does not involve the duplication of genes as in duplication.
 - Can lead to altered gene expression or chromosomal instability.

d) Inversion

- **Inversion** occurs when a chromosome segment breaks off, rotates 180°, and reattaches to the same chromosome.
 - This results in the **reversal of gene loci**, which can affect gene function and lead to various disorders or developmental issues.





Gene Mutation

Gene mutation refers to changes in a specific gene or genetic code at a particular location on a chromosome. These mutations can alter the sequence of nucleotides in DNA, affecting the corresponding proteins and potentially changing an organism's traits.

Types of Gene Mutation

1. Point Mutation

- **Point mutation** involves a change in a single nucleotide in the DNA sequence, which alters the corresponding amino acid in the protein. This can result in:
 - **Non-functional proteins**
 - **Altered protein functions**, which may lead to new traits in an organism.

Examples of Gene Mutation

a) Sickle Cell Anemia

- **Cause:** A **point mutation** at the sixth position of the beta chain of hemoglobin (Hb).
 - The normal code for **glutamic acid** is replaced by **valine**, causing hemoglobin molecules to polymerize.
 - This mutation leads to **sickle-shaped red blood cells** that can block blood flow, causing pain and organ damage.
- **Genotype:**
 - **Normal:** Homozygous HbA HbA.
 - **Affected:** Homozygous HbS HbS.
 - **Carrier (Heterozygous):** HbA HbS (do not experience symptoms, but carry the gene).
- **Symptoms:** Fatigue, pain, fever, fast heartbeat, breathlessness, and organ damage.
- **Effect of Mutation:** The sickle-shaped red blood cells have reduced oxygen-carrying capacity and can accumulate in blood vessels, blocking flow.
- **Treatment:** Blood transfusions, pain relief, increased fluid intake, and **bone marrow transplant** as a long-term solution.

b) Phenylketonuria (PKU)

- **Cause:** A **point mutation** in the gene responsible for producing the enzyme **phenylalanine hydroxylase**, which converts phenylalanine into tyrosine.
 - This mutation causes **phenylalanine** to accumulate in the body, resulting in the production of **phenylketones** that damage the nervous system.
 - PKU appears in **homozygous recessive** individuals.





- **Symptoms:** Severe intellectual disability and nervous system damage if untreated.
- **Treatment:** A **phenylalanine-free diet** (especially during childhood) and the use of **special formula milk** (e.g., Lofenalac) for infants with PKU.

Conclusion

Gene mutations, such as point mutations, can significantly alter protein structure and function, leading to genetic disorders like **Sickle Cell Anemia** and **Phenylketonuria**. Early detection and appropriate treatment, such as dietary adjustments or blood transfusions, can help manage these disorders.

